

## Multiple Choice (10 marks)

1. The quantum efficiency of a photon detector is 0.1. If  $100$  photons are sent into the detector, one after the other, the detector will detect photons
  - (a) an average of 10 times, with an rms deviation of about 4
  - (b) an average of 10 times, with an rms deviation of about 3
  - (c) an average of 10 times, with an rms deviation of about 1
  - (d) an average of 10 times, with an rms deviation of about 0.1
2. A grating spectrometer can just barely resolve two wavelengths of 500 nm and 502 nm, respectively. Which of the following gives the resolving power of the spectrometer?
  - (a) 2
  - (b) 250
  - (c) 5,000
  - (d) 10,000
3. The emission spectrum of the doubly ionized lithium atom  $\text{Li}^{++}$  ( $Z = 3$ ,  $A = 7$ ) is identical to that of a hydrogen atom in which all the wavelengths are
  - (a) decreased by a factor of 9
  - (b) decreased by a factor of 49
  - (c) decreased by a factor of 81
  - (d) increased by a factor of 9
4. The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately
  - (a)  $0.1 \text{ GeV}/c^2$
  - (b)  $0.2 \text{ GeV}/c^2$
  - (c)  $0.5 \text{ GeV}/c^2$
  - (d)  $1.0 \text{ GeV}/c^2$
5. The sign of the charge carriers in a doped semiconductor can be deduced by measuring which of the following properties?
  - (a) Specific heat
  - (b) Thermal conductivity
  - (c) Electrical resistivity
  - (d) Hall coefficient

## True / False (10 marks)

*Answer True or False*

6. True or False: In a constant volume process, the increase in the internal energy of an ideal gas is equal to the heat added to the gas.
7. True or False: In the diamond structure of elemental carbon, the nearest neighbors of each C atom are located at the corners of a tetrahedron.
8. True or False: The total spin quantum number of the electrons in the ground state of neutral nitrogen is  $3/2$ .
9. True or False: Bosons have symmetric wave functions and obey the Pauli exclusion principle.
10. True or False: An atom with filled  $n = 1$  and  $n = 2$  energy levels has a total of 10 electrons.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. A particle has a proper decay time of 2.0 ms. If it travels at  $0.60c$  in the lab frame, calculate the distance it covers before decaying and explain the underlying relativistic effects.
12. Discuss the relationship between the absolute temperature of a blackbody and the energy it radiates per second per unit area, specifically addressing how an increase in temperature affects this energy output.

*End of Paper*